

g : 2D Riemannian 計量, $p \in \Sigma$: 第一種特異点印.

(u, v) : nsacs

$$g = E du^2 + G dv^2$$

$$\tilde{K} = v^2 K = d_0(u) + \underline{d_1(u)} v + d_2(u) v^2 + \dots$$

$$\begin{cases} d_0(u) := \tilde{K}(u, 0) \\ d_1(u) := \frac{\partial}{\partial v} \tilde{K}(u, 0) \end{cases} \quad \text{と定義}$$

問 $d_0(u), d_1(u)$ は不変量か?
(特に, (u, v) のとり方に依らず?)

$f: \Sigma \rightarrow \mathbb{R}^3$: 3-type edge

$$ds_f^2 := f^* \langle \cdot, \cdot \rangle_{\mathbb{R}^3}$$

$$\begin{aligned} \Pi_{3,4}(u) &:= \tilde{K}_v(u) W_{3,4}(u) \\ &= \frac{3}{2} \tilde{K}(u, 0) \\ &= \frac{3}{2} d_0(u) \end{aligned}$$

定理 $\Pi_{3,4}(u)$ は内在的.

問 $\Pi_{3,5}(u)$ と $d_1(u)$ の関係?
 ($\Pi_{3,4}(u) = \text{part}$, $\Pi_{3,5}(u) = \text{定数} \times d_1(u)$)
T.E.3)

$$= v^2 \cdot \tilde{\lambda}(u, v)$$

$$\lambda_v = 2v \cdot \tilde{\lambda} + v^2 \tilde{\lambda}_v$$

$$\lambda_w = 2 \tilde{\lambda}(u, 0) + v \cdot \tilde{\lambda}_v$$

$$= 2F_u(u, 0)$$

$$\lambda_w(u, 0) = 2 \tilde{\lambda}(u, 0)$$

$$= 1.$$

$$u, 0) = 0.$$

$$2F_u = \frac{\lambda^3}{v^2 \times (C^{\infty p} \cdot E(u, v))}$$

$$E_v = v^2 E(u, v)$$

$$E_{vv} = 2v E + v^2 E_v$$

$$E_{vv} = 2E + v \cdot (C^{\infty})$$

$$E_{vv}(u, 0) = 2E(u, 0)$$

g : 2D Kossowski 計量, $p \in \Sigma$: 第一種
 (u, v) : nsacs / (ξ, η) : 別の nsacs 招集点印.

$$g = E du^2 + G dv^2$$

$$\tilde{K} = v^2 K = (\beta(u) + \gamma(u)v + \delta(u, v)v^2)$$

$$\begin{cases} \beta(u) := \tilde{K}(u, 0) \\ \gamma(u) := \frac{\partial}{\partial v} \tilde{K}(u, 0) \end{cases} \text{と定まる}$$

問 $\beta(u), \gamma(u)$ は不変量か?
 (特に, (u, v) のとり方に依らず?)

$f: \Sigma \rightarrow \mathbb{R}^3$: 3-type edge

$$ds_f^2 := f^* \langle \cdot, \cdot \rangle_{\mathbb{R}^3}$$

$$\begin{aligned} \Pi_{3,4}(u) &:= \tilde{K}_v(u) W_{3,4}(u) \\ &= \frac{3}{2} \tilde{K}(u, 0) \\ &= \frac{3}{2} \beta(u) \end{aligned}$$

定理 $\Pi_{3,4}(u)$ は内積.

問 $\Pi_{2,5}(u)$ と $\gamma(u)$ の関係?
 ($\Pi_{2,4}(u) = 0$ かつ, $\Pi_{2,5}(u) = (\text{定数}) \times \gamma(u)$)

$$\lambda^2 = EG - F^2$$

$$= v^4 \times \left(\frac{1}{4} C^{\infty} f_{\mu\nu} \right)$$

$$E(u,v) = 1 + \cancel{E_1(u)v} + \cancel{E_2(u)v^2} + E_3(u)v^3 + E_4(u,v)v^4$$

$$G(u,v) = \frac{1}{4}v^4 + G_5(u)v^5 + G_6(u)v^6 + G_7(u)v^7 + G_8(u,v)v^8$$

計量から定まる不変量
(2つ)

- $K_5(u) := -\frac{1}{2} E_{uv}(u,0)$
- $\beta(u) := \tilde{K}(u,0)$
- $\gamma(u) := \frac{\partial}{\partial v} \tilde{K}(u,0)$

(補題) 定理 3.8

$$K_5(u) = -\frac{E_{uv}(u,0)}{2\sqrt{E(u,0)} \lambda_{\mu\nu}(u,0)}$$

$$= -\frac{E_{uv}(u,0)}{2 \cdot 1 \cdot 1} = -\frac{1}{2} E_{uv}(u,0)$$

$(u,v), (\xi,\eta) : nsacs$

$$E_{\eta\eta}(\xi,0) \stackrel{?}{=} E_{vv}(u,0)$$

$$\frac{\partial}{\partial \eta} = \frac{\partial u}{\partial \eta} \frac{\partial}{\partial u} + \frac{\partial v}{\partial \eta} \frac{\partial}{\partial v}$$

$$\therefore E_{\eta} = u_{\eta} E_u + v_{\eta} E_v$$

$$E_{\eta\eta} = \frac{\partial}{\partial \eta} (u_{\eta} E_u + v_{\eta} E_v)$$

$$= u_{\eta\eta} E_u + u_{\eta} \underbrace{(E_u)_{\eta}} + v_{\eta\eta} E_v + v_{\eta} (E_v)_{\eta}$$

$$= u_{\eta\eta} E_u + u_{\eta} (u_{\eta} E_{uu} + v_{\eta} E_{uv}) + v_{\eta\eta} E_v + v_{\eta} (u_{\eta} E_{uv} + v_{\eta} E_{vv})$$

$$\lambda(u,v) = \lambda_v = 2v \cdot \tilde{\lambda} + v^2 \tilde{\lambda}_v$$

$$\lambda_{uv} = 2 \tilde{\lambda}(u,0) + v \cdot \tilde{C}^{\infty}$$

$$\lambda_{uv}(u,0) = 2 \tilde{\lambda}(u,0) = 1$$

$$E_v(u,0) = 2F_u(u,0)$$

$$E_v(u,0) = 0$$

$$E_v - 2F_u = \frac{\lambda^3}{v^2 \times \tilde{C}^{\infty} f_{\mu\nu}} E(u,v)$$

$$E_v = v^2 E(u,v)$$

$$E_{vv} = 2v E + v^2 E_v$$

$$E_{vv} = 2E + v \cdot \tilde{C}^{\infty}$$

$$E_{vv}(u,0) = 2E(u,0)$$

g : 2x2 Kossowski 計量, $p =$

$(u,v) : nsacs$ / $(\xi,\eta) : \text{別の nsacs}$

$$g = E du^2 + G dv^2$$

$$\tilde{K} = v^2 K = \beta(u) + \gamma(u)$$

$$\begin{cases} \beta(u) := \tilde{K}(u,0) \\ \gamma(u) := \frac{\partial}{\partial v} \tilde{K}(u,0) \end{cases}$$

補題

$\beta(u), \gamma(u)$ は 不変

(例. (u,v) の 11 項)

$$\lambda^2 = EG - F^2$$

$$= v^4 \times \left(\frac{1}{4} C^{\infty} f_{\tau} \right)_{x_0}$$

$$E(u,v) = 1 + \cancel{E_1(u)v} + \cancel{E_2(u)v^2} + E_3(u)v^3 + E_4(u,v)v^4$$

$$G(u,v) = \frac{1}{4}v^4 + G_5(u)v^5 + G_6(u)v^6 + G_7(u)v^7 + G_8(u,v)v^8$$

1. 量 λ の定数下変量
(2.3.3)

- $K_5(u) := -\frac{1}{2} E_{uv}(u,0)$
- $\beta(u) := \tilde{K}(u,0)$
- $\gamma(u) := \frac{\partial}{\partial v} \tilde{K}(u,0)$

(補題) 定理 3.8

$$K_5(u) = \frac{E_{uv}(u,0)}{2\sqrt{E(u,0)} \lambda_{uv}(u,0)}$$

$$= -\frac{E_{uv}(u,0)}{2 \cdot 1 \cdot 1} = -\frac{1}{2} E_{uv}(u,0)$$

$(u,v), (\xi,\eta) : nsacs$

$$E_{\eta\eta\eta}(\xi,0) \stackrel{?}{=} E_{uvv}(u,0)$$

$$\frac{\partial}{\partial \eta} = \frac{\partial u}{\partial \eta} \frac{\partial}{\partial u} + \frac{\partial v}{\partial \eta} \frac{\partial}{\partial v}$$

$$\therefore E_{\eta} = u_{\eta} E_u + v_{\eta} E_v$$

$$E_{\eta\eta} = \frac{\partial}{\partial \eta} (u_{\eta} E_u + v_{\eta} E_v)$$

$$= u_{\eta\eta} E_u + u_{\eta} (E_u)_{\eta} + v_{\eta\eta} E_v + v_{\eta} (E_v)_{\eta}$$

$$= u_{\eta\eta} E_u + u_{\eta} (u_{\eta} E_{uu} + v_{\eta} E_{uv}) + v_{\eta\eta} E_v + v_{\eta} (u_{\eta} E_{uv} + v_{\eta} E_{vv})$$

$$\lambda_{uv} = \tilde{\lambda}(u,v)$$

$$\lambda_v = 2v \cdot \tilde{\lambda} + v^2 \tilde{\lambda}_v$$

$$\lambda_u = 2 \tilde{\lambda}(u,0) + v \cdot \tilde{\lambda}_u$$

$$\lambda_{uv}(u,0) = 2 \tilde{\lambda}_u(u,0) = 1$$

$$E_v(u,0) = 2F_u(u,0)$$

$$E_v(u,0) = 0$$

$$E_v - 2F_u = \frac{\lambda^3}{v^2 \times \left(\frac{1}{4} C^{\infty} f_{\tau} \right)_{x_0}} E(u,v)$$

$$E_{vv} = 2v E_u + v^2 E_{vv}$$

$$E_{vvv} = 2E_u + v \cdot \tilde{C}^{\infty}$$

$$E_{uvv}(u,0) = 2E_u(u,0)$$

g : 2.7 Kossowski の量

$(u,v) : nsacs$ / $(\xi,\eta) : \text{Bil}$

$$g = E du^2 + G dv^2$$

$$\tilde{K} = v^2 K = \beta(u) + \gamma(u)v$$

$$\begin{cases} \rho(u) := \tilde{K}(u,0) \\ \gamma(u) := \frac{\partial}{\partial v} \tilde{K}(u,0) \end{cases}$$

$\beta(u), \gamma(u)$ は $\tilde{K}$ の係数

(例. (u,v) の場合)

$$\lambda^2 = EG - F^2$$

$$= v^4 \times \left(\frac{1}{4} C^{\infty} f_{\alpha\beta} \right)$$

$$\lambda = v^2 \left(\frac{1}{2} C^{\infty} f_{\alpha\beta} \right)$$

$$\lambda(u,v) = v^2 \cdot \tilde{\lambda}(u,v)$$

$$\lambda_v = \frac{\partial}{\partial v} (v^2 \tilde{\lambda}) = 2v \tilde{\lambda} + v^2 \tilde{\lambda}_v$$

$$\lambda_v(u,0) = 2 \tilde{\lambda}(u,0) = 2$$

$$\tilde{\lambda}_v(u,0) = 1$$

$$E_v(u,0) = 2F_u(u,0)$$

$$E_v(u,0) = 0$$

$$E_v - 2F_u = \frac{\lambda^3}{v^2 \times \left(\frac{1}{4} C^{\infty} f_{\alpha\beta} \right)}$$

$$E_v = v^2 \Sigma(u,v)$$

$$E_{vv} = 2v \Sigma + v^2 \Sigma_v$$

$$E_{vv} = 2\Sigma + v \cdot \left(\frac{1}{4} C^{\infty} \right)$$

$$E_{vv}(u,0) = 2 \Sigma(u,0)$$

$(u,v), (\xi,\eta) : nsacs$

$$E_{\eta\eta}(\xi,0) \stackrel{?}{=} E_{vv}(u,0)$$

$$\frac{\partial}{\partial \eta} = \frac{\partial u}{\partial \eta} \frac{\partial}{\partial u} + \frac{\partial v}{\partial \eta} \frac{\partial}{\partial v}$$

$$\therefore E_{\eta} = u_{\eta} E_u + v_{\eta} E_v$$

$$E_{\eta\eta} = \frac{\partial}{\partial \eta} (u_{\eta} E_u + v_{\eta} E_v)$$

$$= u_{\eta\eta} E_u + u_{\eta} \left(\frac{\partial E_u}{\partial \eta} \right) + v_{\eta\eta} E_v + v_{\eta} \left(\frac{\partial E_v}{\partial \eta} \right)$$

$$= u_{\eta\eta} E_u + u_{\eta} (u_{\eta} E_{uu} + v_{\eta} E_{uv}) + v_{\eta\eta} E_v + v_{\eta} (u_{\eta} E_{uv} + v_{\eta} E_{vv})$$

g : 2x2 Kossowski 計量, $p \in \Sigma$ 第一種
 $(u,v) : nsacs$ / $(\xi,\eta) : \text{別の nsacs}$ 特異点印

$$g = E du^2 + G dv^2$$

$$\tilde{K} = v^2 K = (\beta(u) + \gamma(u)v + \delta(u,v)v^2)$$

$$\left\{ \begin{array}{l} \rho(u) := \tilde{K}(u,0) \\ \gamma(u) := \frac{\partial}{\partial v} \tilde{K}(u,0) \end{array} \right. \text{と定めた}$$

$\rho(u), \gamma(u)$ は 不変量か?
 (特異点 (u,v) の u,v 方に依存しない?)

$$f: \Sigma \rightarrow \mathbb{R}^3$$

$$ds_f^2 := f^* \langle \cdot, \cdot \rangle$$

$$\Pi_{3,4}(u) = \tilde{K}$$

$$= \frac{3}{2}$$

$$= \frac{3}{2}$$

定理 $\Pi_{3,4}(u)$

$$\Pi_{2,5}(u)$$

$$(\Pi_{3,4}(u))$$

$$\lambda^2 = E^2 - P^2 = v^2 \cdot (E^2 - P^2) = v^2 \cdot (E^2 - P^2)$$

$$E(u,v) = 1 + E(u,v) + E(u,v)^2 + E(u,v)^3 + E(u,v)^4$$

$$G(u,v) = \frac{1}{4} v^4 + G(u,v) v^2 + G(u,v) v^2 + G(u,v) v^2 + G(u,v) v^2$$

1. 证明下列变量
(2.3.3.3)

$$K_s(u) = -\frac{1}{2} E_{uv}(u,0)$$

$$\beta(u) = \tilde{K}(u,0)$$

$$\gamma(u) = \frac{\partial}{\partial v} \tilde{K}(u,0)$$

(证明) 定理 18

$$K_s(u) = \frac{E_{uv}(u,0)}{2 \sqrt{E(u,0)} \lambda_{uv}(u,0)}$$

$$= -\frac{E_{uv}(u,0)}{2 \cdot 1 \cdot 1} = -\frac{1}{2} E_{uv}(u,0)$$

$(u,v), (s,t): nsacs$

$$E_{\eta\eta\eta}(s,0) \stackrel{?}{=} E_{uv}(u,0)$$

$$\frac{\partial}{\partial \eta} = \frac{\partial u}{\partial \eta} \frac{\partial}{\partial u} + \frac{\partial v}{\partial \eta} \frac{\partial}{\partial v}$$

$$E_{\eta} = u_{\eta} E_u + v_{\eta} E_v$$

$$E_{\eta\eta} = \frac{\partial}{\partial \eta} (u_{\eta} E_u + v_{\eta} E_v)$$

$$= u_{\eta\eta} E_u + u_{\eta} (E_u)_{\eta} + v_{\eta\eta} E_v + v_{\eta} (E_v)_{\eta}$$

$$= u_{\eta\eta} E_u + u_{\eta} (u_{\eta} E_{uu} + v_{\eta} E_{uv}) + v_{\eta\eta} E_v + v_{\eta} (u_{\eta} E_{uv} + v_{\eta} E_{vv})$$

$$\lambda(u,v) = v^2$$

$$E_v(u,0) = 2(u,0)$$

$$E_v(u,0) = 0$$

$$E_v - 2E_u = \frac{\lambda^3}{v^2 \cdot (E^2 - P^2)}$$

$$E_v = v^2 E(u,v)$$

$$E_{vv} = 2v E + v^2 E_v$$

$$E_{vv} = 2E + v \cdot (E^2 - P^2)$$

$$E_{vv}(u,0) = 2E(u,0)$$

$$g = E du^2 + G dv^2 = E$$

$$\begin{cases} du = u_{\eta} ds + u_{\eta} d\eta \\ dv = v_{\eta} ds + v_{\eta} d\eta \end{cases}$$

$$E du^2 + G dv^2 = E (u_{\eta}^2 ds^2 + \dots)$$

$$= \frac{E u_{\eta}^2}{\tilde{E}} + \dots$$

$$\tilde{E} = E u_{\eta}^2 + G v_{\eta}^2 - 0$$

$$0 = E u_{\eta} u_{\eta} + G v_{\eta} v_{\eta} - 0$$

$$\tilde{G} = E u_{\eta}^2 + G v_{\eta}^2 - 0$$

$$E(u,0) = \tilde{E}(s,0) = 1$$

$$G(u,0) = G_v(u,0) = G_{vv}(u,0)$$

$$\tilde{G}(s,0) = \tilde{G}_{\eta}(s,0) = \tilde{G}_{\eta\eta}(s,0)$$

$$\lambda^2 = E^2 G - F^2$$

$$= v^4 \times \left(\frac{1}{4} C^{\infty} f_{\alpha\beta} \right)_{x_0}$$

$$+ E(u,v) v^3 + E_4(u,v) v^4$$

$$(u,v)^6 + G_7(u,v) v^7 + G_8(u,v) v^8$$

$(u,v), (\xi,\eta) : \eta$ sacs

$$E_{\eta\eta\eta}(\xi,0) \stackrel{?}{=} E_{vvv}(u,0)$$

$$\frac{\partial}{\partial \eta} = \frac{\partial u}{\partial \eta} \frac{\partial}{\partial u} + \frac{\partial v}{\partial \eta} \frac{\partial}{\partial v}$$

$$\therefore E_{\eta} = u_{\eta} E_u + v_{\eta} E_v$$

$$E_{\eta\eta} = \frac{\partial}{\partial \eta} (u_{\eta} E_u + v_{\eta} E_v)$$

$$= u_{\eta\eta} E_u + u_{\eta} \underbrace{(E_u)_{\eta}} + v_{\eta\eta} E_v + v_{\eta} (E_v)_{\eta}$$

$$= u_{\eta\eta} E_u + u_{\eta} (u_{\eta} E_{uu} + v_{\eta} E_{uv}) + v_{\eta\eta} E_v + v_{\eta} (u_{\eta} E_{uv} + v_{\eta} E_{vv})$$

$$\lambda(u,v) = v^2 \cdot \tilde{\lambda}(u,v)$$

$$\lambda_v = 2v \cdot \tilde{\lambda}(u,v)$$

$$\lambda_w = 2 \tilde{\lambda}(u,0) \cdot v \times C^{\infty}$$

$$\lambda_w(u,0) = 2 \tilde{\lambda}(u,0) = 1$$

$$E_v(u,0) = 2 F_u(u,0)$$

$$E_v(u,0) = 0$$

$$E_v - 2 F_u = \frac{\lambda^3}{v^2} \times \left(C^{\infty} f_{\alpha\beta} \right)_{x_0}$$

$$E_v = v^2 \mathcal{E}(u,v)$$

$$E_{vv} = 2v \mathcal{E} + v^2 E_v$$

$$E_{vv} = 2\mathcal{E} + v \cdot C^{\infty}$$

$$E_{vv}(u,0) = 2 \mathcal{E}(u,0)$$

$$g = E du^2 + G dv^2 = \tilde{E} d\xi^2 + \tilde{G} d\eta^2$$

$$\begin{cases} du = u_{\xi} d\xi + u_{\eta} d\eta \\ dv = v_{\xi} d\xi + v_{\eta} d\eta \end{cases}$$

$$E du^2 + G dv^2 = E (u_{\xi} d\xi + u_{\eta} d\eta)^2 + G (v_{\xi} d\xi + v_{\eta} d\eta)^2$$

$$= \underbrace{E u_{\xi}^2 + G v_{\xi}^2}_{\tilde{E}} d\xi^2 + 2 (E u_{\xi} u_{\eta} + G v_{\xi} v_{\eta}) d\xi d\eta + \underbrace{(E u_{\eta}^2 + G v_{\eta}^2)}_{\tilde{G}} d\eta^2$$

$$\therefore \begin{cases} \tilde{E} = E u_{\xi}^2 + G v_{\xi}^2 - ① \\ 0 = E u_{\xi} u_{\eta} + G v_{\xi} v_{\eta} - ② \\ \tilde{G} = E u_{\eta}^2 + G v_{\eta}^2 - ③ \end{cases}$$

$$E(u,0) = \tilde{E}(\xi,0) = 1 \quad \sim E_v(u,0) = E_w(u,0) = 0, \quad \tilde{E}_{\eta}(u,0) = \tilde{E}_{\eta\eta}(u,0) = 0$$

$$G(u,0) = G_v(u,0) = G_w(u,0) = G_{vv}(u,0) = 0, \quad G_{ww}(u,0) = 6$$

$$\tilde{G}(\xi,0) = \tilde{G}_{\eta}(\xi,0) = \tilde{G}_{\eta\eta}(\xi,0) = \tilde{G}_{\eta\eta\eta}(\xi,0) = 0, \quad \tilde{G}_{\eta\eta\eta\eta}(\xi,0) = 6$$

① $|_{v=0}$

② $|_{v=0}$

③ $|_{v=0}$

$\frac{\partial}{\partial \eta}$ ①

$$\lambda(u,v) = v^2 \cdot \tilde{\lambda}(u,v)$$

$$\lambda_v = 2v \cdot \tilde{\lambda} + v^2 \tilde{\lambda}_v$$

$$\lambda_u = 2 \tilde{\lambda}(u,0) + v^2 \tilde{\lambda}_u(u,0)$$

$$\lambda(u,0) = 2 \tilde{\lambda}(u,0)$$

$$\tilde{\lambda}_u(u,0) = 1$$

$$E_v(u,0) = 0$$

$$E_v - 2\tilde{F}_u = \frac{\lambda^3}{v^2}$$

$$E_v = v^2 \mathcal{E}(u,v)$$

$$E_{vv} = 2v \mathcal{E} + v^2 \mathcal{E}_v$$

$$E_{uv} = 2\mathcal{E} + v \mathcal{E}_u$$

$$E_{uv}(u,0) = 2\mathcal{E}(u,0)$$

$$g = E du^2 + G dv^2 = \tilde{E} d\xi^2 + \tilde{G} d\eta^2$$

$$\begin{cases} du = u_\xi d\xi + u_\eta d\eta \\ dv = v_\xi d\xi + v_\eta d\eta \end{cases}$$

$$E du^2 + G dv^2 = E (u_\xi d\xi + u_\eta d\eta)^2 + G (v_\xi d\xi + v_\eta d\eta)^2$$

$$= \underbrace{E u_\xi^2 + G v_\xi^2}_{\tilde{E}} d\xi^2 + \underbrace{2(E u_\xi u_\eta + G v_\xi v_\eta)}_{\tilde{G}} d\xi d\eta + (E u_\eta^2 + G v_\eta^2) d\eta^2$$

$$\begin{cases} \tilde{E} = E u_\xi^2 + G v_\xi^2 - 1 & (1) \\ 0 = E u_\xi u_\eta + G v_\xi v_\eta & (2) \\ \tilde{G} = E u_\eta^2 + G v_\eta^2 & (3) \end{cases}$$

$$E(u,0) = \tilde{E}(\xi,0) = 1 \quad E_v(u,0) = \tilde{E}_v(u,0) = 0, \quad \tilde{E}_\eta(u,0) = \tilde{E}_{\eta\eta}(u,0) = 0$$

$$G(u,0) = G_v(u,0) = G_w(u,0) = G_{vv}(u,0) = 0, \quad G_{uv}(u,0) = 6$$

$$\tilde{G}(\xi,0) = \tilde{G}_\eta(\xi,0) = \tilde{G}_{\eta\eta}(\xi,0) = \tilde{G}_{\eta\eta\eta}(\xi,0) = 0, \quad \tilde{G}_{\eta\eta\eta\eta}(\xi,0) = 6$$

$$1|_{v=0}; \quad 1 = 1 \cdot u_\xi(\xi,0)^2 \quad \therefore u_\xi(\xi,0) = \pm 1$$

$$\left(\frac{\partial(u,v)}{\partial(\xi,\eta)} \right) = \begin{vmatrix} u_\xi & u_\eta \\ v_\xi & v_\eta \end{vmatrix} \stackrel{v=0}{=} u_\xi v_\eta > 0$$

$$2|_{v=0}; \quad 0 = 1 \cdot (\pm 1) u_\eta + 0 \quad \therefore u_\eta(\xi,0) = 0$$

$$3|_{v=0}; \quad 0 = 0$$

$$\frac{\partial}{\partial \eta} (1); \quad \tilde{E}_\eta = E_\eta u_\xi^2 + E(u_\xi^2)_\eta + G_\eta v_\xi^2 + G(v_\xi^2)_\eta$$

$$\tilde{E}_\eta = (u_\eta E_v + v_\eta E_v) u_\xi^2 + E \cdot 2 u_\xi u_{\xi\eta} + (u_\eta G_u + v_\eta G_u) v_\xi^2 + G \cdot (v_\xi^2)_\eta$$

$$\eta=0; \quad 0 = 0 + 0 = 0$$

$$\tilde{E}_{\eta\eta}(\xi,0) = \tilde{E}_{\eta\eta\eta}(\xi,0) = 0$$

$$\tilde{E}_{\eta\eta\eta}(\xi,0) = \tilde{E}_{\eta\eta\eta\eta}(\xi,0) = 0$$

$$\lambda^2 = EG - F^2 = v^4 \times \left(\frac{1}{4} C^{\infty} f_{\tau} \right)_{x_0}$$

$$E(u,v) = 1 + \cancel{E_1(u)}v + \cancel{E_2(u)}v^2 + E_3(u)v^3 + E_4(u,v)v^4$$

$$G(u,v) = \frac{1}{4}v^4 + G_5(u)v^5 + G_6(u)v^6 + G_7(u)v^7 + G_8(u,v)v^8$$

計量から定まる変数
(2つ)

$$K_s(u) := -\frac{1}{2} E_{uv}(u,0)$$

$$\beta(u) := \tilde{K}(u,0)$$

$$\gamma(u) := \frac{\partial}{\partial v} \tilde{K}(u,0)$$

(福岡) 定理 3.8

$$K_s(u) = -\frac{E_{uv}(u,0)}{2\sqrt{E(u,0)} \lambda_{uv}(u,0)}$$

$$= -\frac{E_{uv}(u,0)}{2 \cdot 1 \cdot 1} = -\frac{1}{2} E_{uv}(u,0)$$

$(u,v), (\xi,\eta) : \eta$ sacs

$$\hat{E}_{\eta\eta}(\xi,0) \stackrel{?}{=} E_{vv}(u,0)$$

$$\frac{\partial}{\partial \eta} = \frac{\partial u}{\partial \eta} \frac{\partial}{\partial u} + \frac{\partial v}{\partial \eta} \frac{\partial}{\partial v}$$

$$\therefore E_{\eta} = u_{\eta} E_u + v_{\eta} E_v$$

$$E_{\eta\eta} = \frac{\partial}{\partial \eta} (u_{\eta} E_u + v_{\eta} E_v)$$

$$= u_{\eta\eta} E_u + u_{\eta} \underbrace{(E_u)_{\eta}} + v_{\eta\eta} E_v + v_{\eta} (E_v)_{\eta}$$

$$= u_{\eta\eta} E_u + u_{\eta} (u_{\eta} E_{uu} + v_{\eta} E_{uv}) + v_{\eta\eta} E_v + v_{\eta} (u_{\eta} E_{uv} + v_{\eta} E_{vv})$$

$$\lambda_v = 2v \cdot \tilde{\lambda} + v^2 \tilde{\lambda}_v$$

$$\lambda_{uv} = 2 \tilde{\lambda}(u,0) + v \cdot \tilde{C}^{\infty}$$

$$\lambda_{uv}(u,0) = 2 \tilde{\lambda}(u,0) = 1$$

$$E_v(u,0) = 2F_u(u,0)$$

$$E_v(u,0) = 0$$

$$E_v - 2F_u = \frac{\lambda^3}{v^2 \times \tilde{C}^{\infty} f_{\tau}} E(u,v)$$

$$E_v = v^2 E(u,v)$$

$$E_{vv} = 2v E + v^2 E_v$$

$$E_{vv} = 2E + v \cdot \tilde{C}^{\infty}$$

$$E_{vv}(u,0) = 2E(u,0)$$

$$g = E du^2 + G dv^2$$

$$\begin{cases} du = u_{\eta} \\ dv = v_{\eta} \end{cases}$$

$$E du^2 + G dv^2$$

$$\begin{cases} \tilde{E} = \\ 0 = \\ \tilde{G} = \end{cases}$$

$$E_{uv}$$

$$G(u,v)$$

$$\tilde{G}^{\infty}$$

$$= 2v \cdot \tilde{\lambda} + v^2 \tilde{\lambda}_v$$

$$\lambda_w = 2 \tilde{\lambda}(u, 0) + v \cdot \tilde{\lambda}_v$$

$$\lambda_w(u, 0) = 2 \tilde{\lambda}(u, 0)$$

$$= 1.$$

$$\exists \frac{1}{2} \times (C^{\infty} \rho + \dots)$$

$$\Sigma(u, v)$$

$$v \Sigma + v^2 \Sigma_v$$

$$\Sigma_{uv} = 2 \Sigma + v \cdot C^{\infty}$$

$$\Sigma_{uv}(u, 0) = 2 \Sigma(u, 0)$$

$$v_\eta (E_\eta)_\eta$$

$$v_\eta E_\eta + v_\eta (u_\eta E_{uv} + v_\eta E_{vv})$$

$$g = E du^2 + G dv^2 = \hat{E} d\xi^2 + \hat{G} d\eta^2$$

$$\begin{cases} du = u_\xi d\xi + u_\eta d\eta \\ dv = v_\xi d\xi + v_\eta d\eta \end{cases}$$

$$E du^2 + G dv^2 = E (u_\xi d\xi + u_\eta d\eta)^2 + G (v_\xi d\xi + v_\eta d\eta)^2$$

$$= \underbrace{E u_\xi^2 + G v_\xi^2}_{\hat{E}} d\xi^2 + 2 (E u_\xi u_\eta + G v_\xi v_\eta) d\xi d\eta + \underbrace{E u_\eta^2 + G v_\eta^2}_{\hat{G}} d\eta^2$$

$$\begin{cases} \hat{E} = E u_\xi^2 + G v_\xi^2 - ① \\ 0 = E u_\xi u_\eta + G v_\xi v_\eta - ② \\ \hat{G} = E u_\eta^2 + G v_\eta^2 - ③ \end{cases}$$

$$E(u, 0) = \hat{E}(\xi, 0) = 1 \quad E_v(u, 0) = E_w(u, 0) = 0, \quad \hat{E}_\eta(u, 0) = \hat{E}_w(u, 0) = 0$$

$$G(u, 0) = G_v(u, 0) = G_w(u, 0) = 0, \quad G_{vv}(u, 0) = 6, \quad G_{uv}(u, 0) = 6$$

$$\hat{G}(\xi, 0) = \hat{G}_\eta(\xi, 0) = \hat{G}_{\eta\eta}(\xi, 0) = \hat{G}_{\eta w}(\xi, 0) = 0, \quad \hat{G}_{\eta\eta\eta}(\xi, 0) = 6$$

$$①|_{v=0}; \quad 1 = 1 \cdot u_\xi(\xi, 0)^2 \quad \therefore u_\xi(\xi, 0) = \pm 1$$

$$\left(\left| \frac{\partial(u, v)}{\partial(\xi, \eta)} \right| = \begin{vmatrix} u_\xi & u_\eta \\ v_\xi & v_\eta \end{vmatrix} \right) \stackrel{v=0}{=} u_\xi v_\eta > 0$$

$$②|_{v=0}; \quad 0 = 1 \cdot (\pm 1) \cdot u_\eta + 0 \quad \therefore u_\eta(\xi, 0) = 0$$

$$③|_{v=0}; \quad 0 = 0$$

$$\frac{\partial}{\partial \eta} ①; \quad \hat{E}_\eta = E_\eta u_\xi^2 + E(u_\xi^2)_\eta + G_\eta v_\xi^2 + G(v_\xi^2)_\eta$$

$$\hat{E}_\eta = \underbrace{(u_\eta E_v + v_\eta E_v)}_{\dots} u_\xi^2 + E \cdot 2 u_\xi u_{\xi\eta} + \underbrace{(u_\eta G_v + v_\eta G_v)}_{\dots} v_\xi^2 + G \cdot (v_\xi^2)_\eta$$

$$\eta=0; \quad 0 = 0 + \dots$$

$$\hat{E}_{\eta\eta}(\xi, 0) = \hat{E}_{\eta\eta}(\xi, 0) = \dots$$